

Web Programming

Overview of web development

Instructors

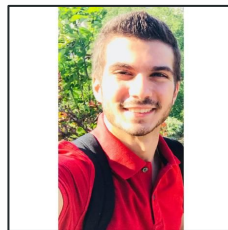


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Bilal EL CHAMI

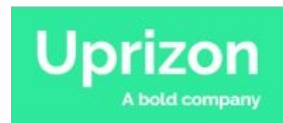
Tech Lead @ Uprizon

bilalchamil@gmail.com

github.com/bilal-elchami



**SOCIETE
GENERALE**



Present yourself

- Alternance / Internship Company (*if applicable*)
- Career projection
- Experience in web development (*Java, Spring Boot, Angular et Git*)

Technical stack

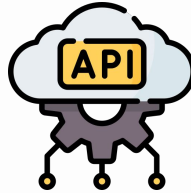
Database



PostgreSQL



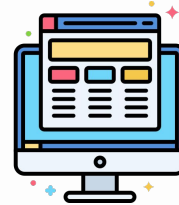
Backend



Java 25
Spring Boot 4



Frontend



Typescript
Angular 21



Sessions

formation classique

Session #01	<i>23 January</i>	Overview of Web development	
Session #02	<i>30 January</i>	Building a frontend application with Angular	
Session #03	<i>06 February</i>	Building a backend application with Spring Boot	
Session #04	<i>13 February</i>	Implementing RESTful API	
Session #05	<i>20 February</i>	Hands on frontend	
Session #06	<i>27 February</i>	Hands on frontend	
Session #07	<i>20 March</i>	TP Review + Final Project focus	 
Session #08	<i>27 March</i>	Final Project Presentations	 

Sessions

formation alternance

Session #01	<i>30 January</i>	Overview of Web development	
Session #02	<i>12 February</i>	Building a frontend application with Angular	
Session #03	<i>19 February</i>	Building a backend application with Spring Boot	
Session #04	<i>12 March</i>	Implementing RESTful API	
Session #05	<i>26 March</i>	Hands on frontend	
Session #06	<i>10 April</i>	Hands on frontend	
Session #07	<i>16 April</i>	TP Review + Final Project focus	 
Session #08	<i>05 June</i>	Final Project Presentations	 

Grading

formation classique

20%



TP

Building a blogging platform



05/03/2025

80%



Final project

Building another web app



26/03/2025



27/03/2025

Resources



Google Drive

All slides can be
found on this [link](#)



tinyurl.com/web-dauphine

Grading

formation alternance

20%



TP

Building a blogging platform



17/04/2025

80%



Final project

Building another web app



04/06/2025



05/06/2025

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found on this [link](#)



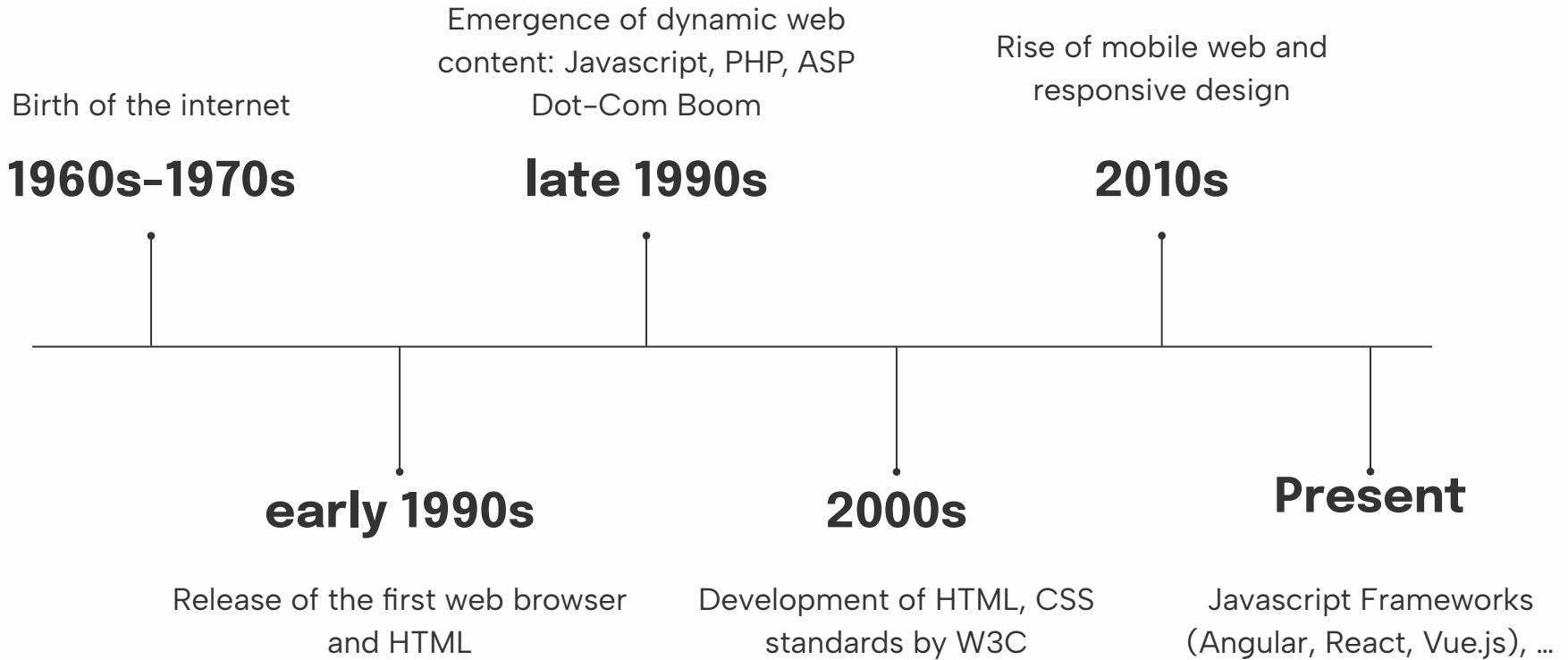
tinyurl.com/web-dauphine

01

Architecture

Some key milestones in web development and evolution of web architecture

Web's milestones



Static website

In the early days of the web, websites were primarily static, consisting of **HTML** files with basic styling using **CSS**.

Content was **hardcoded** into HTML files, and updating content required **manual editing of the HTML**.

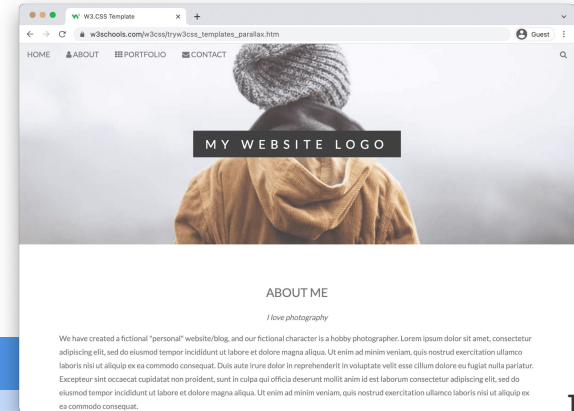
Technical stack : HTML, CSS

Use cases : Informational websites, landing pages, ...

Static website architecture

HTML & CSS

Application



Dynamic website with Database

Server-side scripts allowed websites to generate **dynamic content** by **querying databases** for data and rendering it on the fly.

This led to the rise of dynamic web applications where users could **interact with content, submit forms**, and receive personalized responses.

Technical stack: HTML, CSS, PHP, ASP, JSP

Dynamic website architecture

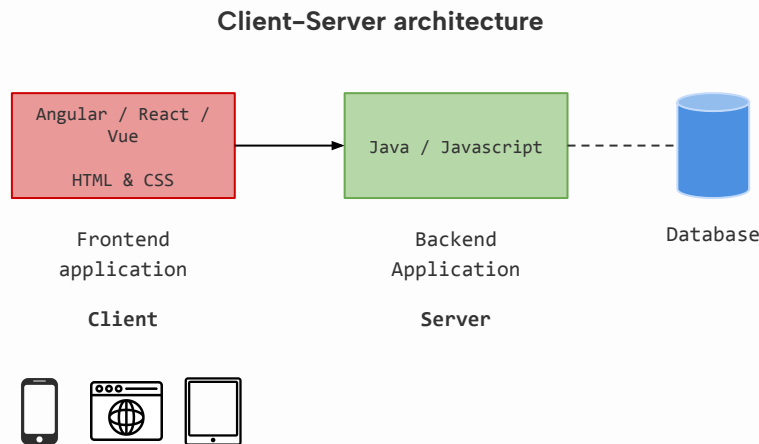


Client-Server (Monolithic)

AJAX (Asynchronous JavaScript and XML) revolutionized web development by enabling **asynchronous communication** between the client and server.

This allowed for more interactive and responsive web applications where **data could be fetched from the server** without reloading the entire page.

Client-side scripting languages like **JavaScript** became more prominent, enabling richer user experiences.

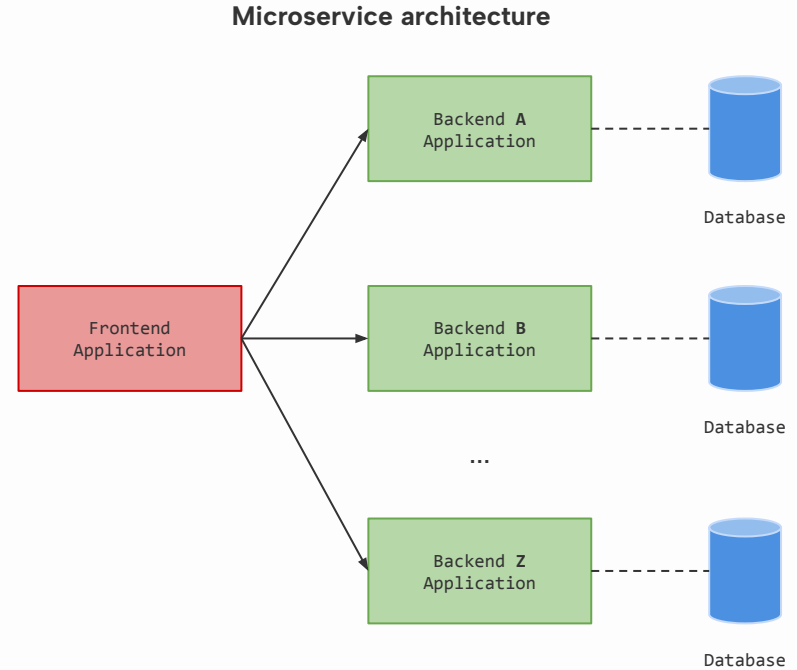


Microservice architecture

As web applications grew in **complexity** and **scale**, the monolithic architecture became less suitable for maintaining and scaling applications.

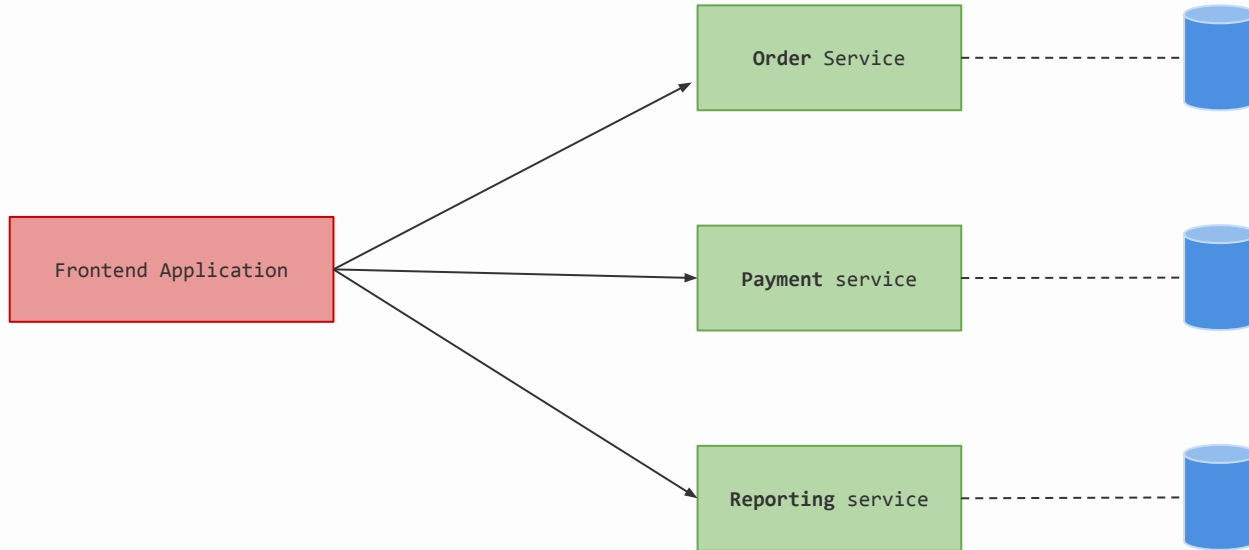
Microservices architecture emerged as an alternative approach, **breaking down applications into smaller, loosely coupled services** that can be developed, deployed, and scaled independently.

Each microservice **focuses on a specific business** function and **communicates** with others **through APIs**



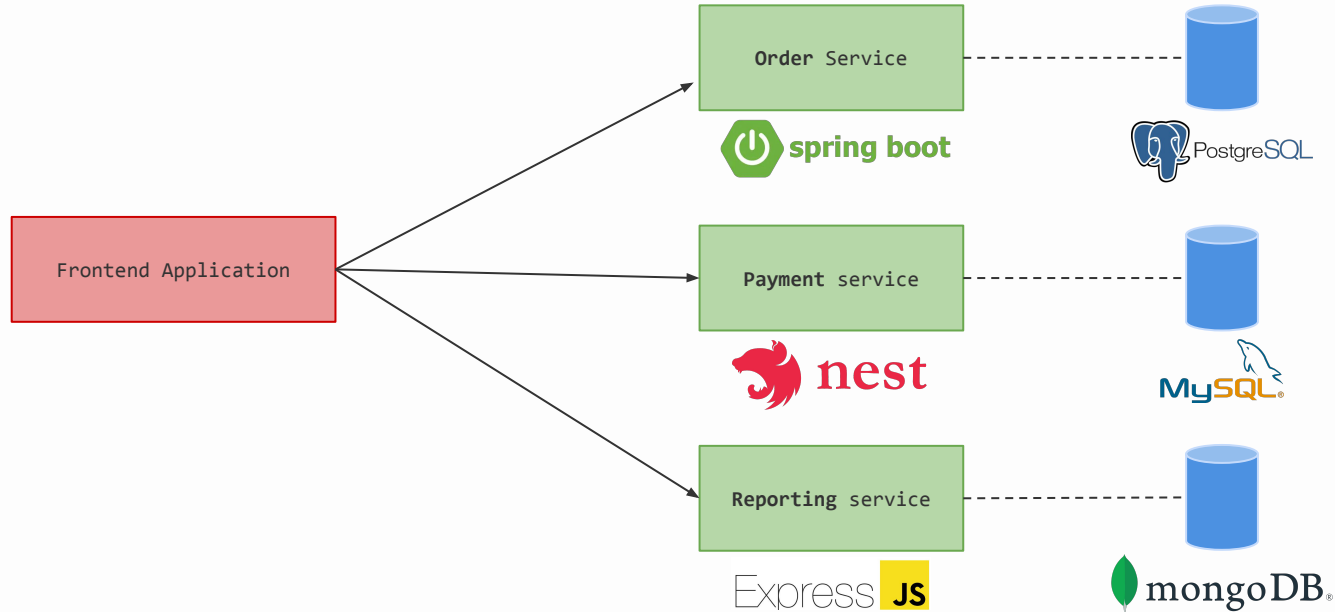
Microservice architecture

Each microservice focuses on a **specific business function**



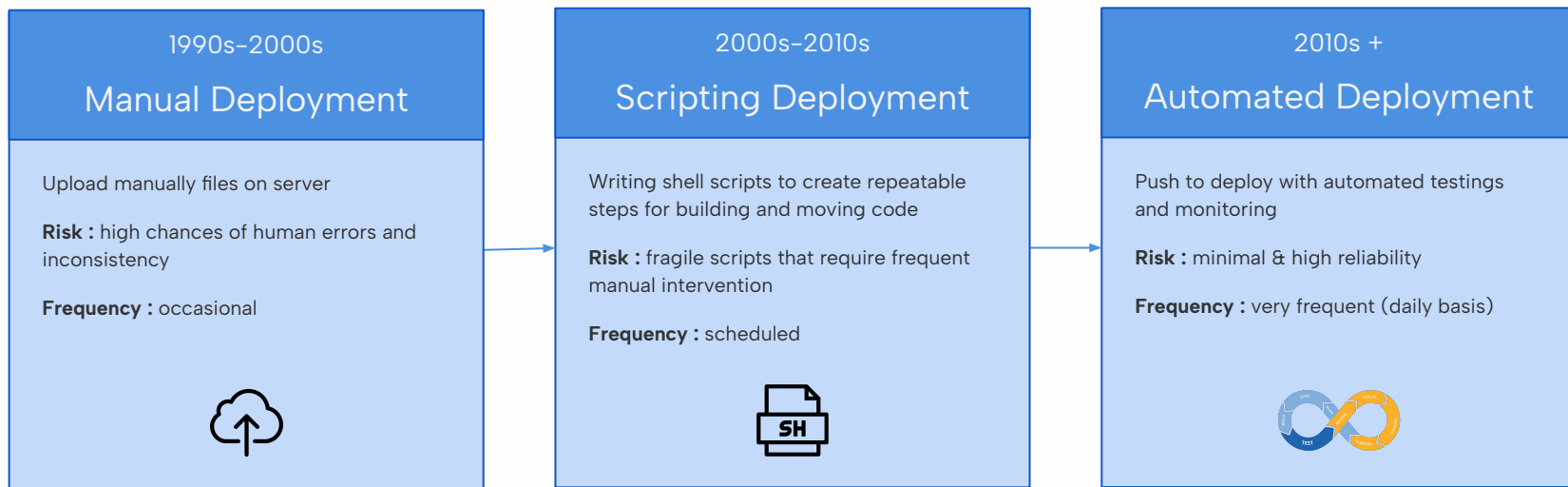
Microservice architecture

It allow us to break down applications into **smaller** and **independent** unit



Evolution of Delivery Practices

During the evolution of web architecture, the way we build, ship and operate applications also changed



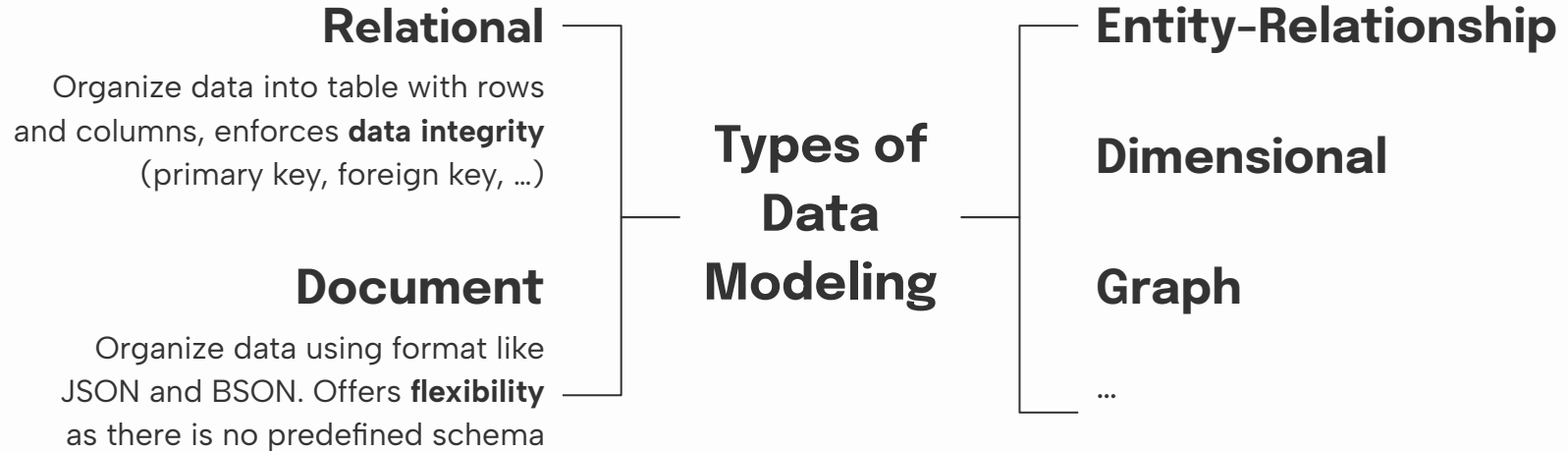
02

Data Modeling

Relational data modeling

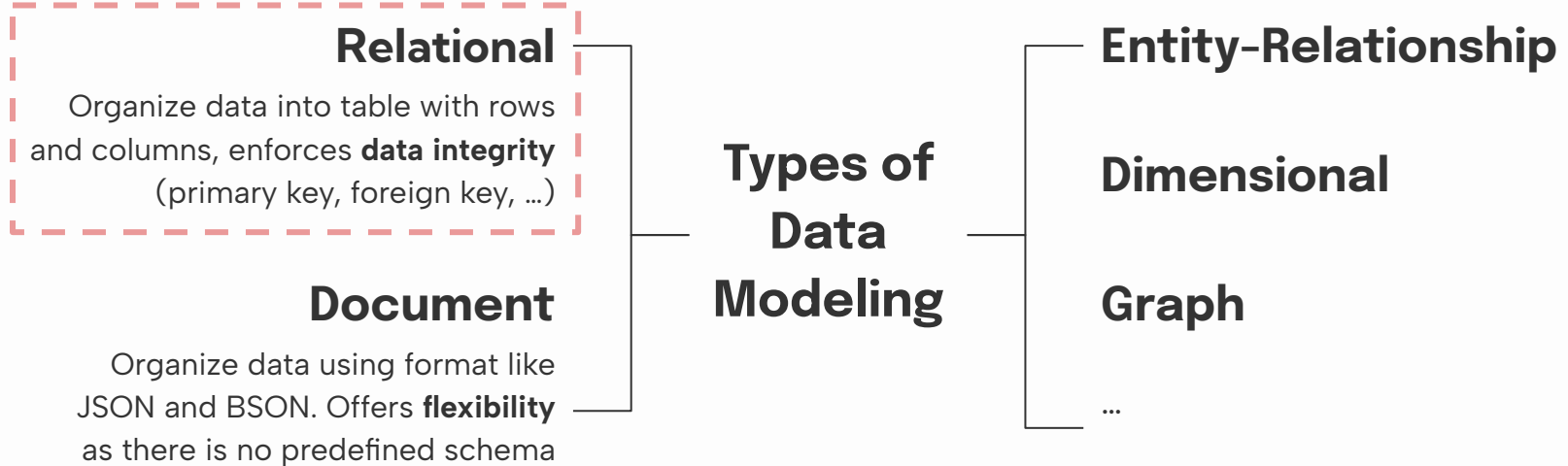
What is a data model ?

A collection of **metadata** that represents the logical organization, shape and meaning of data



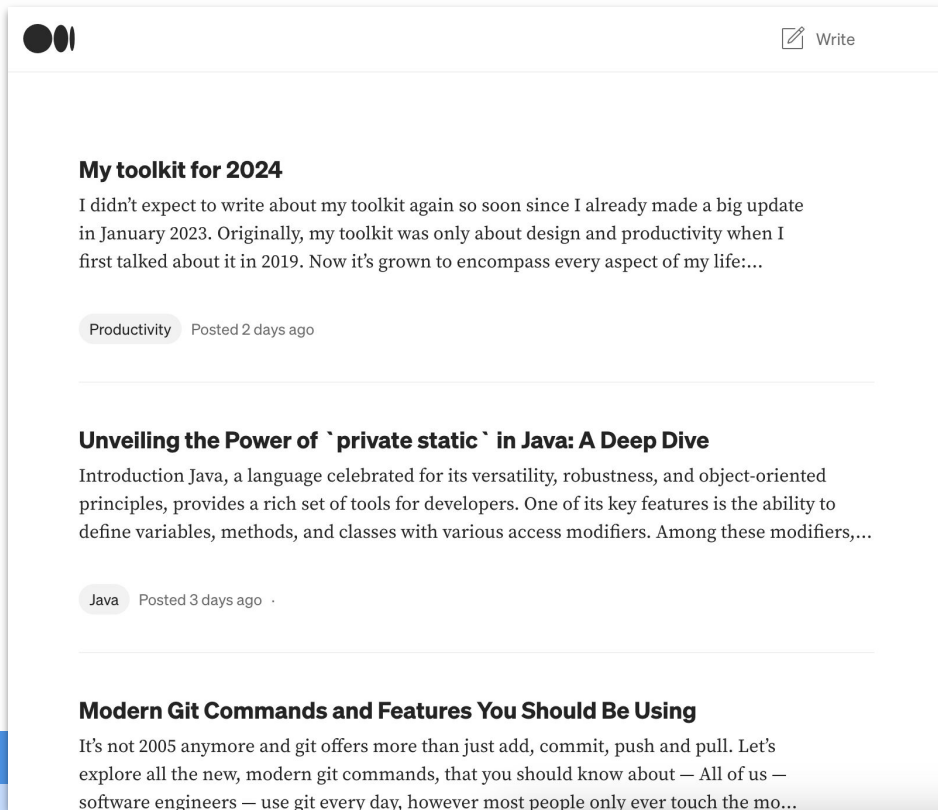
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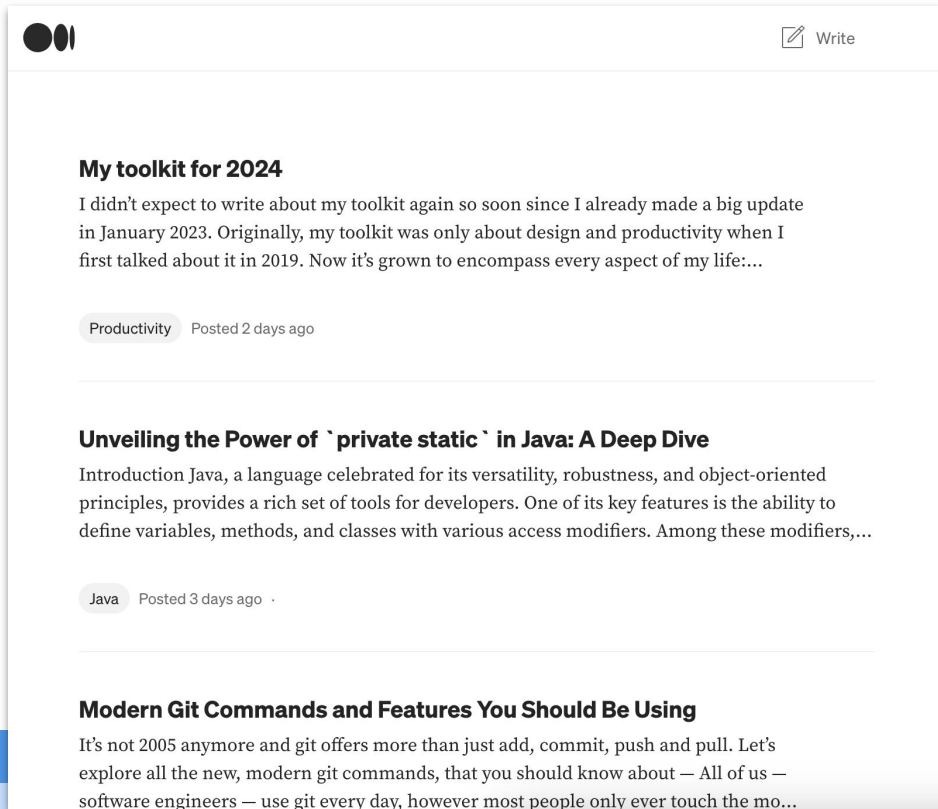


Use case 1 : Blogging platform

Modelisation of a blogging platform

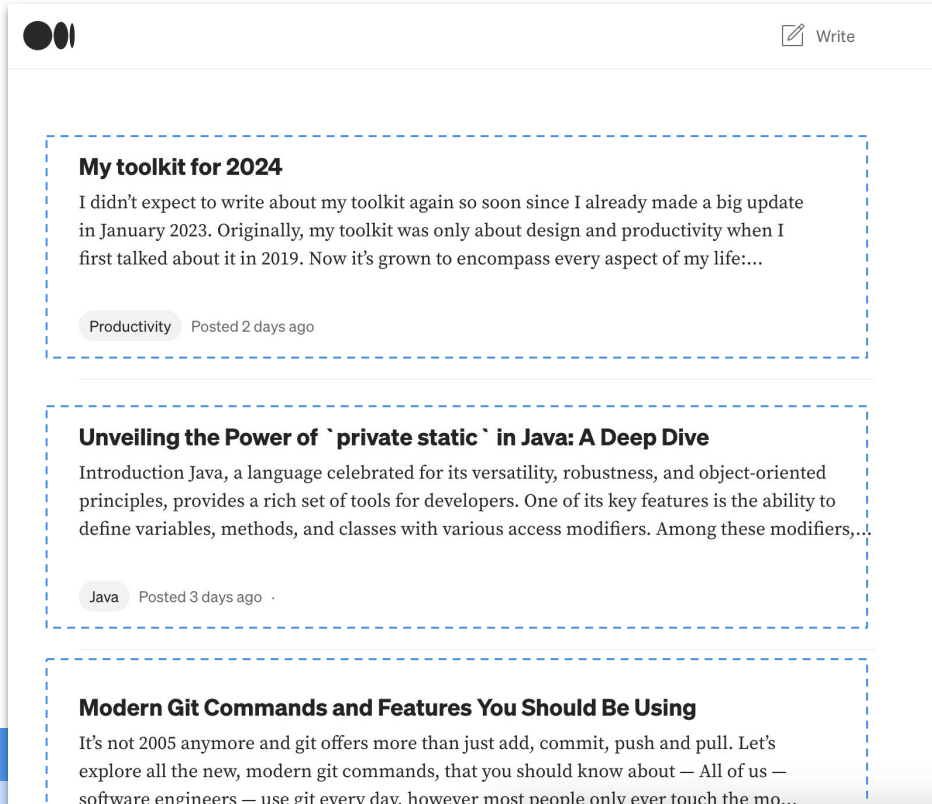


Modelisation of a blogging platform



A blogging platform allows individuals to publish content in the form of posts

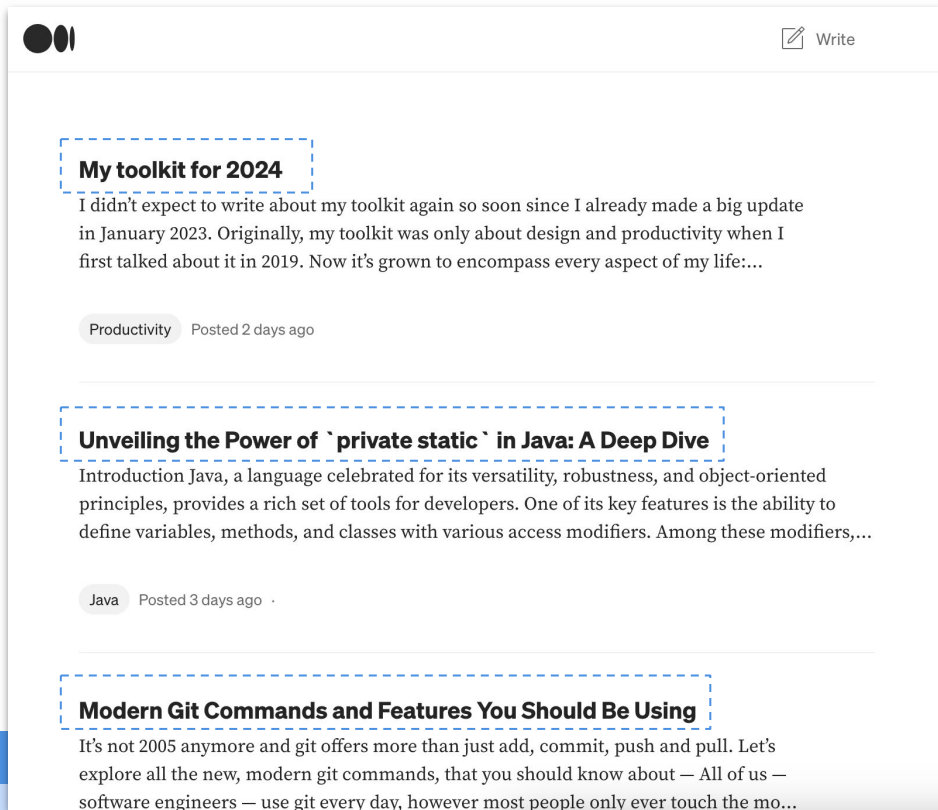
Modelisation of a blogging platform



A blogging platform allows individuals to publish content in the form of posts

Each Post have a

Modelisation of a blogging platform

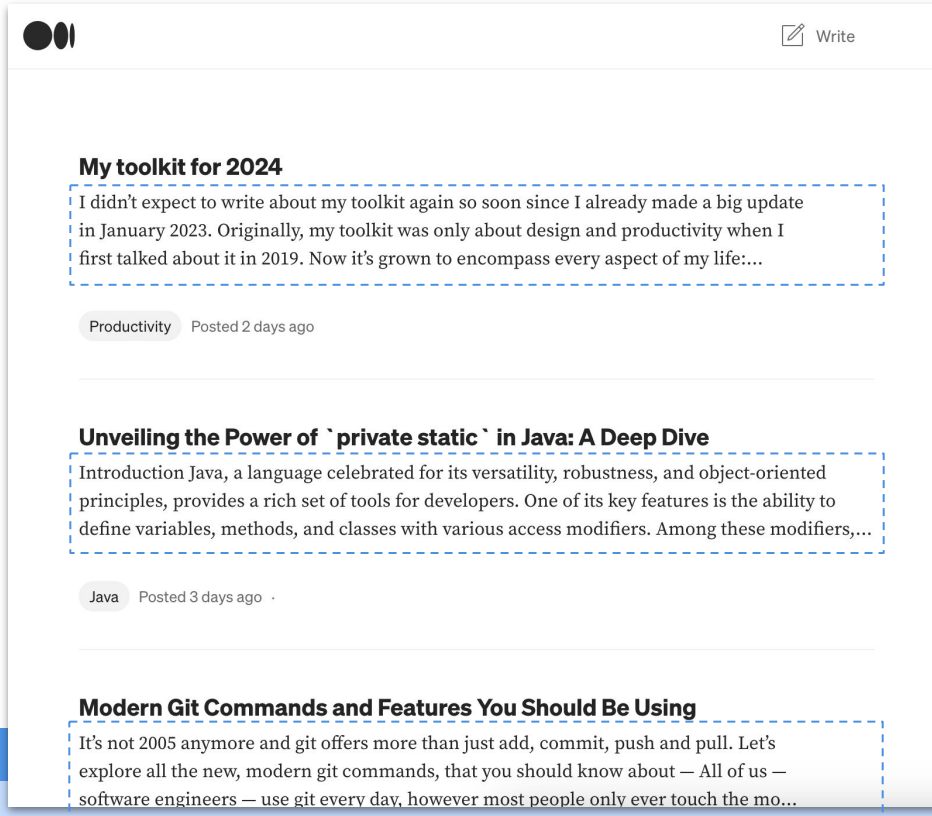


A blogging platform allows individuals to publish content in the form of posts

Each Post have a

- Title

Modelisation of a blogging platform

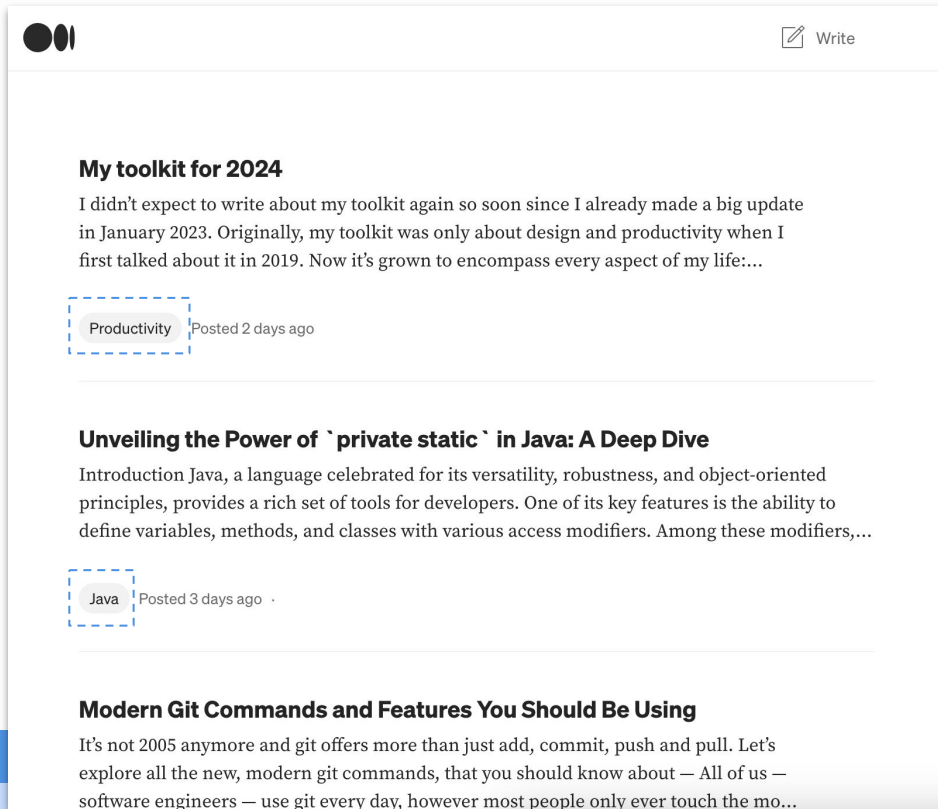


A blogging platform allows individuals to publish content in the form of posts

Each Post have a

- Title
- Content

Modelisation of a blogging platform

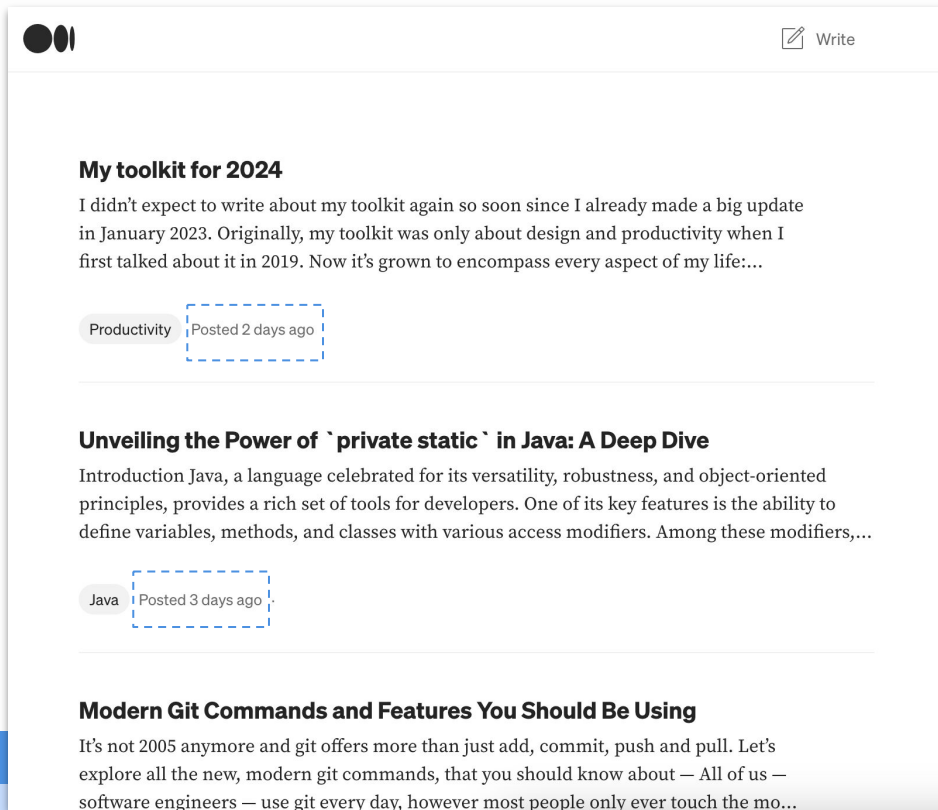


A blogging platform allows individuals to publish content in the form of posts

Each Post have a

- Title
- Content
- Category

Modelisation of a blogging platform



A blogging platform allows individuals to publish content in the form of posts

Each Post have a

- Title
- Content
- Category
- Creation date

Modelisation of a blogging platform

post				
id	title	content	created_date	category
1	My toolkit for 2024	I didn't expect to write about my toolkit again so soon since I already made a big update in January 2023. Originally, my toolkit was ...	31/03/2024	Productivity
2	Unveiling the Power of `private static` in Java: A Deep Dive	Java, a language celebrated for its versatility, robustness, and object-oriented principles, provides a rich set of tools for developers. ...	30/03/2024	Java

Modelisation of a blogging platform

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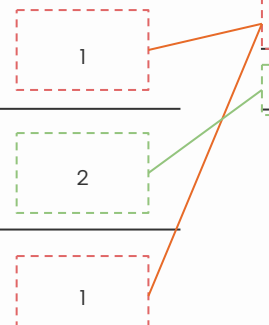
What will happen when we add another new post with the same category ?

Modelisation of a blogging platform

post				
id	title	content	created_date	category
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2	Unveiling the Power of `private static` in Java: A Deep Dive	Java, a language celebrated for its versatility, robustness, and object-oriented principles, provides a rich set of tools for developers. ...	30/03/2024	Java
3	Understand and Overcome Mental Fatigue	Do you ever feel spacey, distracted and worn down toward the end of a long work-related task — especially if that task is entirely a menta l...	29/03/2024	Productivity

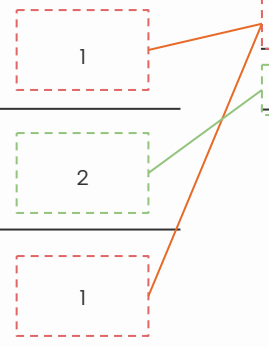
Modelisation of a blogging platform

post					category	
id	title	content	created_date	category_id	id	name
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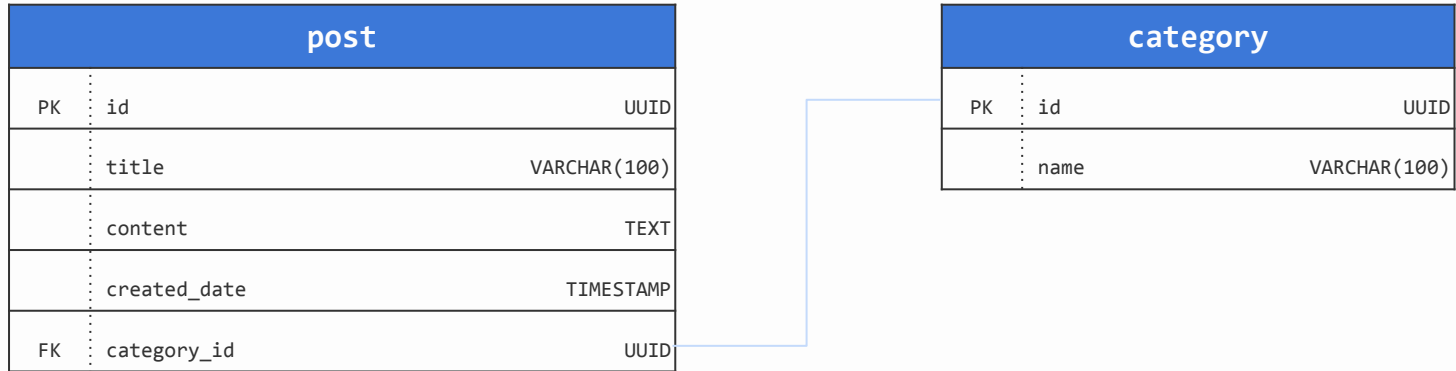
Modelisation of a blogging platform

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Having a foreign key and a category table ensure **data consistency** across the database

Database diagram



You can use diagrams.net or mermaid.live to model it digitally

Use case 2 : Library

Designing and modeling a **library** management system.

The library has **books**, **authors** and **borrowers**.

Use case 3 : Music streaming service

Designing and modeling for a **music streaming service**.

The service should support **songs, playlists, album, artists** and a **playlist** belongs to a **user**.

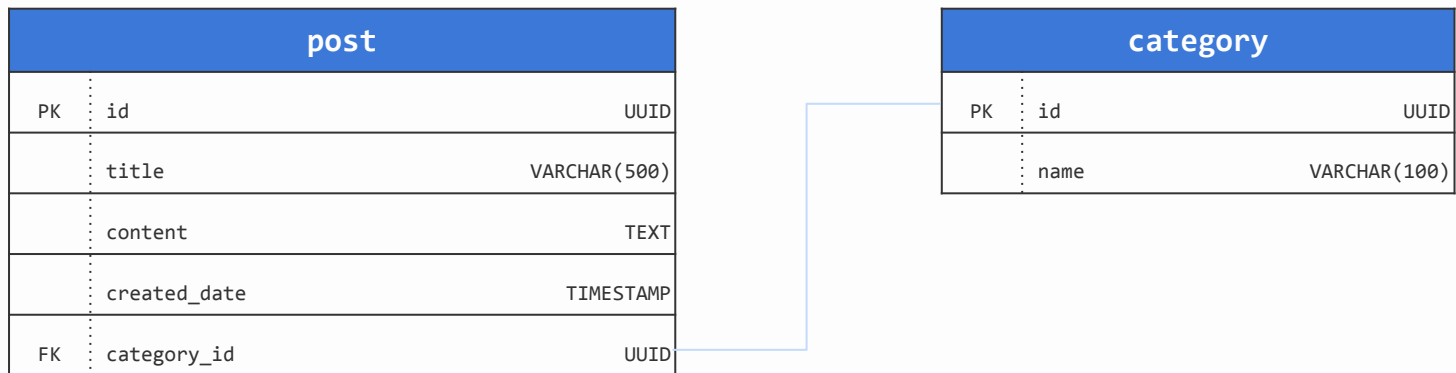
Use case 4 : Marketplace

Designing and modeling a **marketplace**.

The marketplace allows users to **buy** and **sell products**.

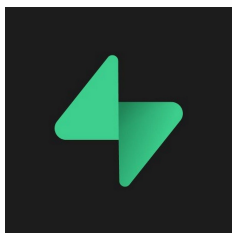
It will also allows support of multiple **categories** for each products and also ability to add a **review** on each product.

Database diagram of a blogging platform



Remote PostgreSQL database

Create and **Connect** to a remotely **PostgreSQL** database via **Supabase**



Supabase





Remote PostgreSQL DB

Head over to supabase.com to create your PostgreSQL database remotely

1. Sign up
2. Create a new organization
3. Create a new project

Project name	blogger-dauphine
Database password	< generate a password >
Region	Europe

Create a new project

Your project will have its own dedicated instance and full Postgres database.
An API will be set up so you can easily interact with your new database.

Organization

devt

FREE



Project name

blogger-dauphine

Database password

••••••••••••••••



Copy

This password is strong. [Generate a password.](#)

Region



Europe



Select the region closest to your users for the best performance.

SECURITY OPTIONS >

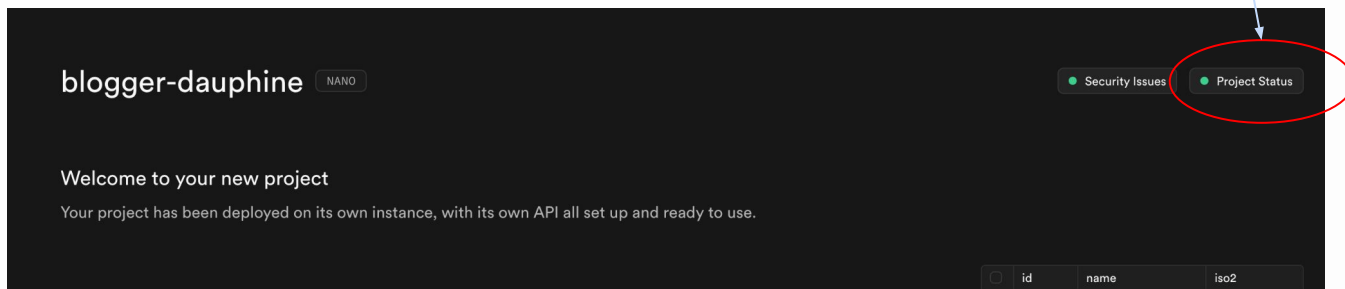
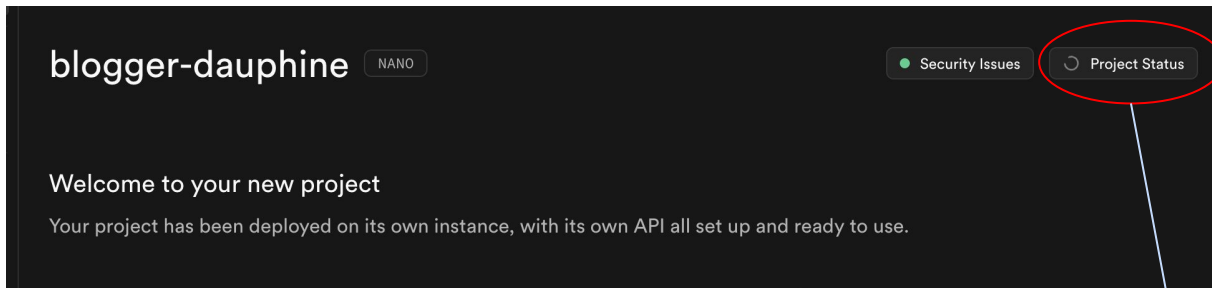
ADVANCED CONFIGURATION >

Cancel

Create new project

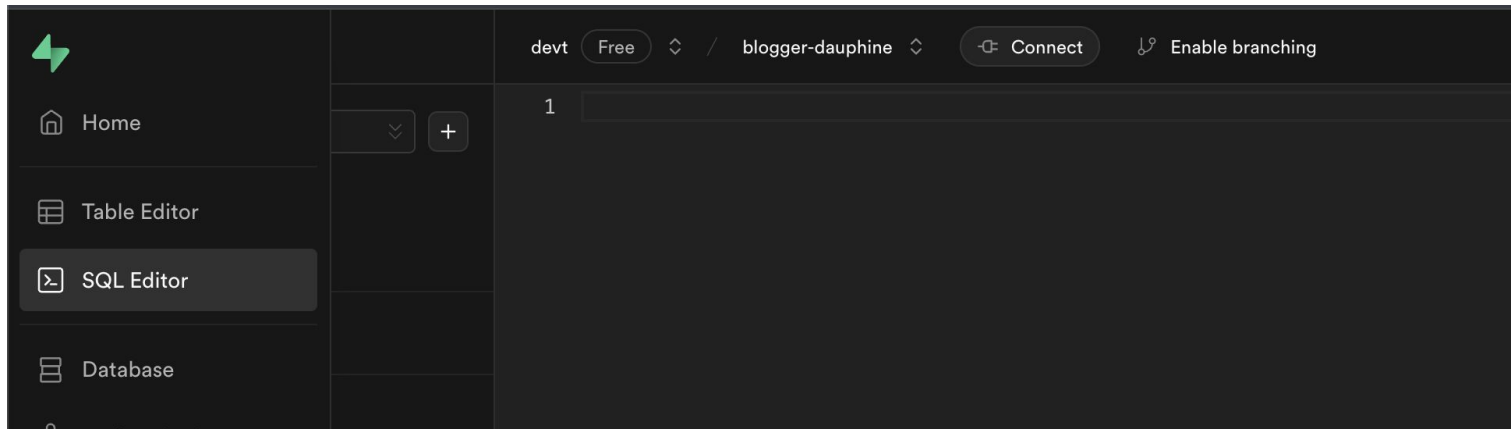


Wait until the Database is ready





Populate blogger database



Use Supabase SQL Editor to create category and post tables



Modelisation in PostgreSQL

Create **category** table

```
CREATE TABLE category (  
  id          UUID,  
  name        VARCHAR(100),  
  PRIMARY KEY (id),  
  UNIQUE(name)  
);
```

Create **post** table

```
CREATE TABLE post (  
  id          UUID,  
  title        VARCHAR(500),  
  content      TEXT,  
  created_date TIMESTAMP,  
  category_id  UUID,  
  PRIMARY KEY (id),  
  FOREIGN KEY (category_id) REFERENCES category(id)  
);
```



Populate data

```
INSERT INTO category (id, name)
VALUES (gen_random_uuid(), 'Productivity'),
       (gen_random_uuid(), 'Java');

INSERT INTO post (id, title, content, created_date, category_id)
VALUES (
    gen_random_uuid(),
    'Title',
    'Content',
    '2023-10-28 11:30:30',
    (SELECT id FROM category WHERE name = 'Productivity')
);
```



Scripts

Script to **create** the tables (Category and Post)

 [create-tables.sql](#)

Script to **populate** data

 [populate-data.sql](#)

03

Version control with Git & GitHub

Getting started with Git and GitHub

What is version control

Version control is a system that **records changes** to files over time so that you can recall specific versions later.

Industry standards

 **git** has become the industry standard for version control, widely adopted by organizations and developers worldwide

Importance of a version control



Collaboration

Multiple developers can work on the same project simultaneously without interfering with each other's work



Branching and merging

Allow developers to create branches for feature development or bug fixes and merge changes into the main codebase



History tracking

Every change made to files is recorded, enabling easy retrieval of previous versions



Backup and recovery

Repositories act as backups, reducing the risk of losing code due to accidents or hardware failures

Git repository hostings



Owned by Microsoft, one of the largest and most popular git repository hosting



Owned by Atlassian, provides integration with Jira and Confluence



Provided by Amazon Web Services, provides integrations other AWS services



Offer entire DevOps lifecycle, issue tracking, code review, ...



Offers cloud-hosted options, with CI/CD pipelines and agile project management, ...

Git repository hostings



GitHub

Owned by Microsoft, one of the largest and most popular git repository hosting



Bitbucket

Owned by Atlassian, provides integration with Jira and Confluence



AWS CodeCommit

Provided by Amazon Web Services, provides integrations other AWS services



GitLab

Offer entire DevOps lifecycle, issue tracking, code review, ...



Azure DevOps

Offers cloud-hosted options, with CI/CD pipelines and agile project management, ...

How Git works

Working directory



Staging Area



Local Repository



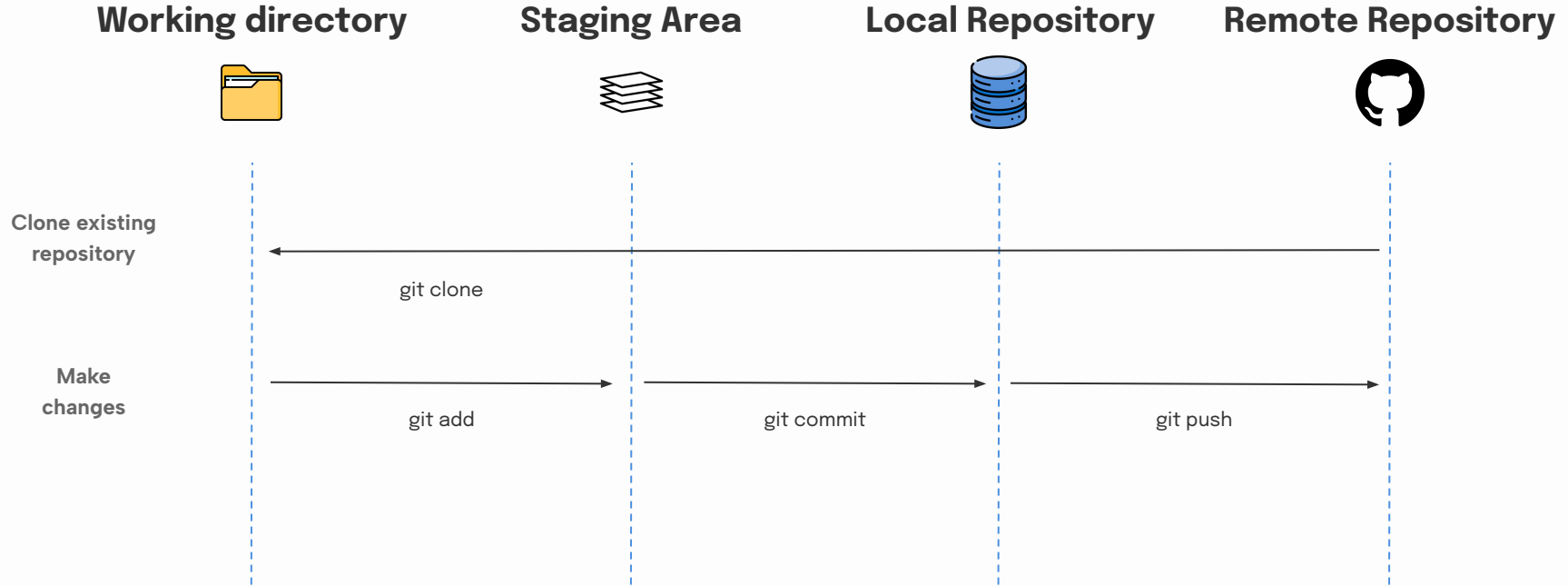
Remote Repository



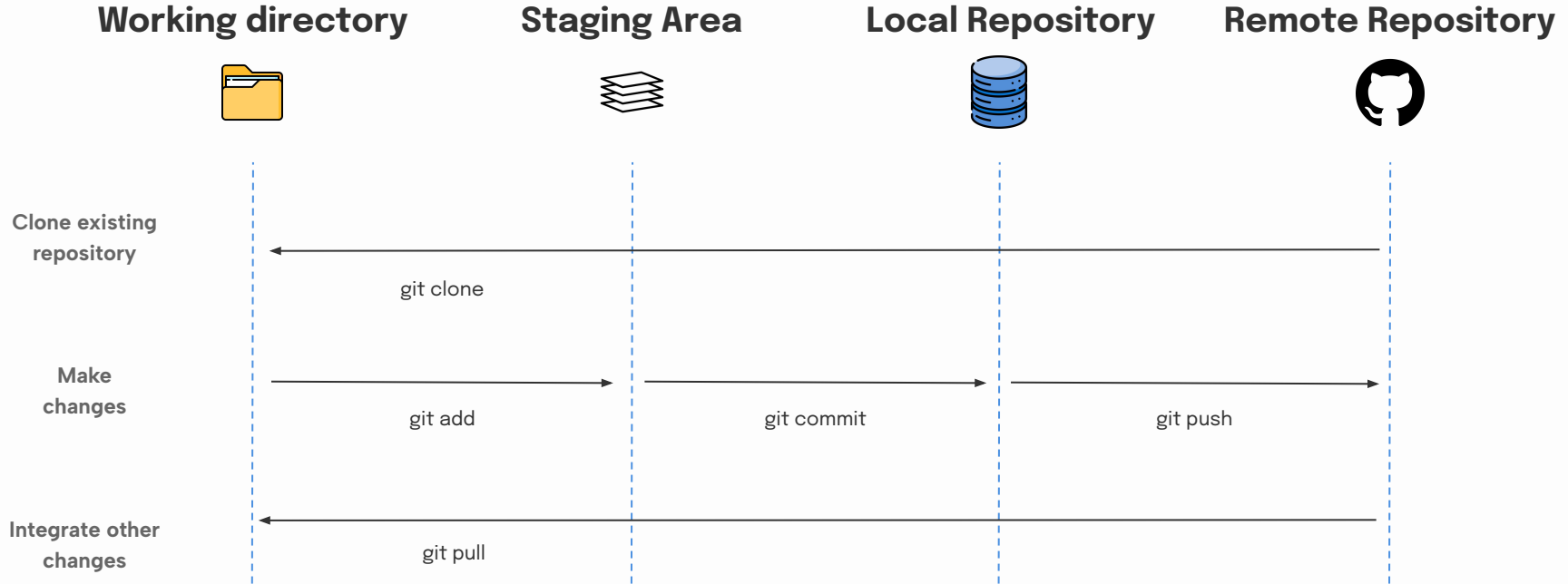
How Git works



How Git works



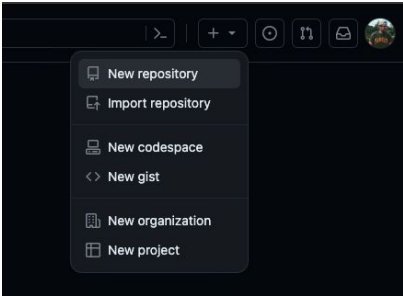
How Git works





Getting started with Git and GitHub

Steps to **create** and **connect** to a remotely **repository on Github**

Step 1	Login to <u>GitHub</u> Follow this documentation to sign up if you don't have an account
Step 2	Create new repository Follow this documentation to learn more  github.com/new 



Getting started with GitHub

Steps to **create** and **connect** to a remotely **repository** on Github

Step 3	Clone project locally, by running command <code>git clone https://github.com/USERNAME/REPOSITORY_NAME</code>
Step 4	Make a changes aka code...
Step 5	Commit and push changes into remotely repository <code>git add .</code> <code>git commit -m "COMMIT_MESSAGE"</code> <code>git push origin BRANCH_NAME</code>

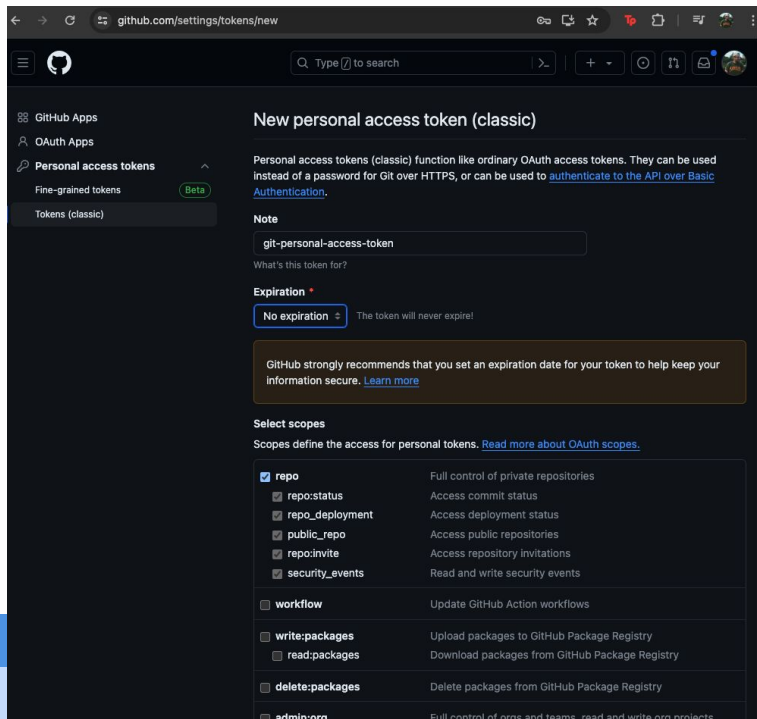


Getting started with GitHub

You will be prompted to add your **username** and **password**.

For the password you will have to
Generate a classical personal access token
with “repo” as scope

 new github token





Getting started with GitHub

Steps to **pull** changes from a remote Github **repository**

Step 1	Fetch references from remote repository <code>git fetch</code>
Step 2	Pull changes <code>git pull</code>



[Git Cheat Sheet](#)

Feeling lazy ?

If you don't feel like learning git command line, download **Github Desktop** which provide a graphical user interface for the Git version control system.

It allows users to perform common **Git operations** such as committing, branching, merging, and syncing repositories.

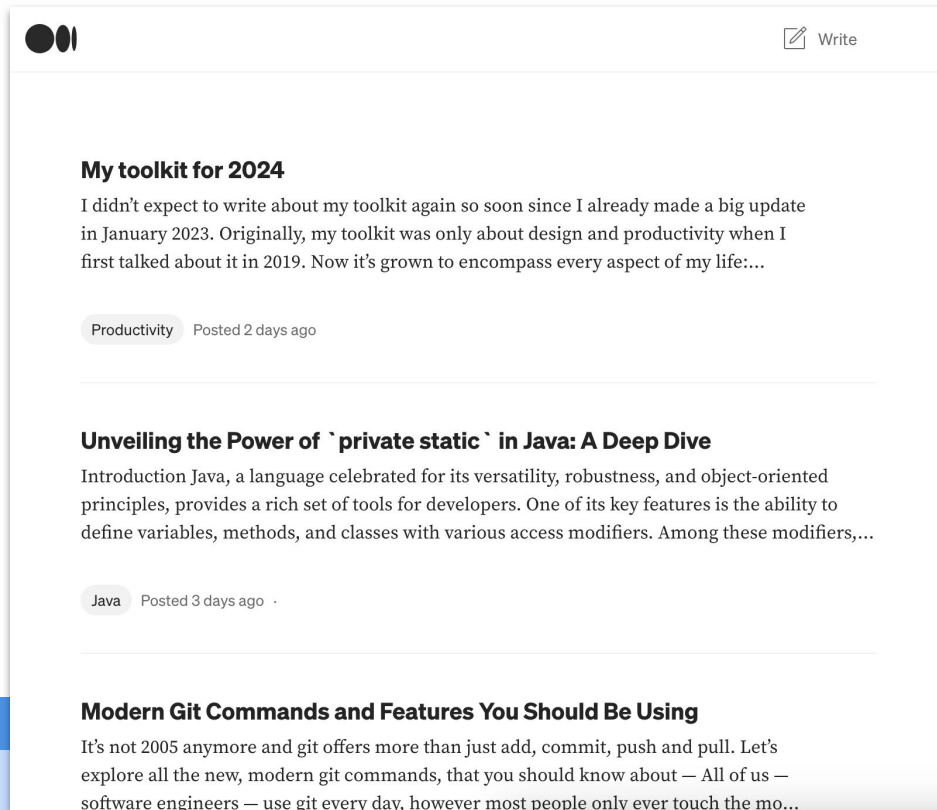


Github Desktop

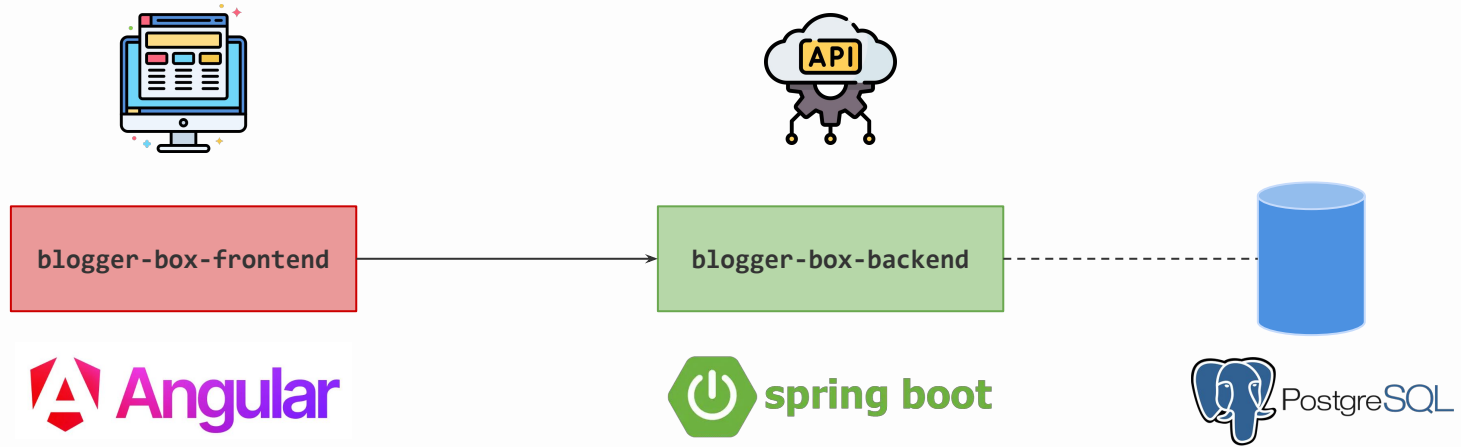
04

Backend / Frontend

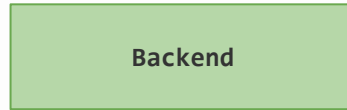
Blogger box



Blogger box architecture



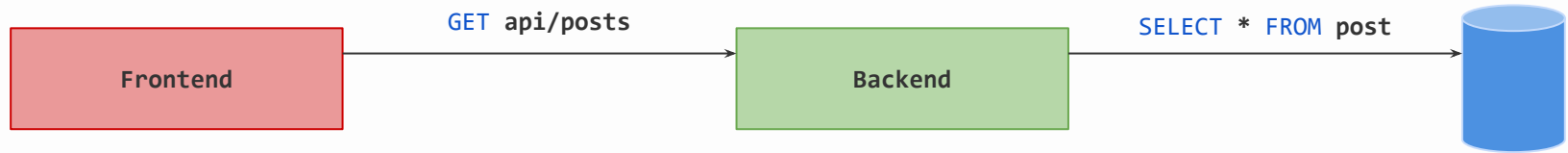
Flow frontend <-> backend



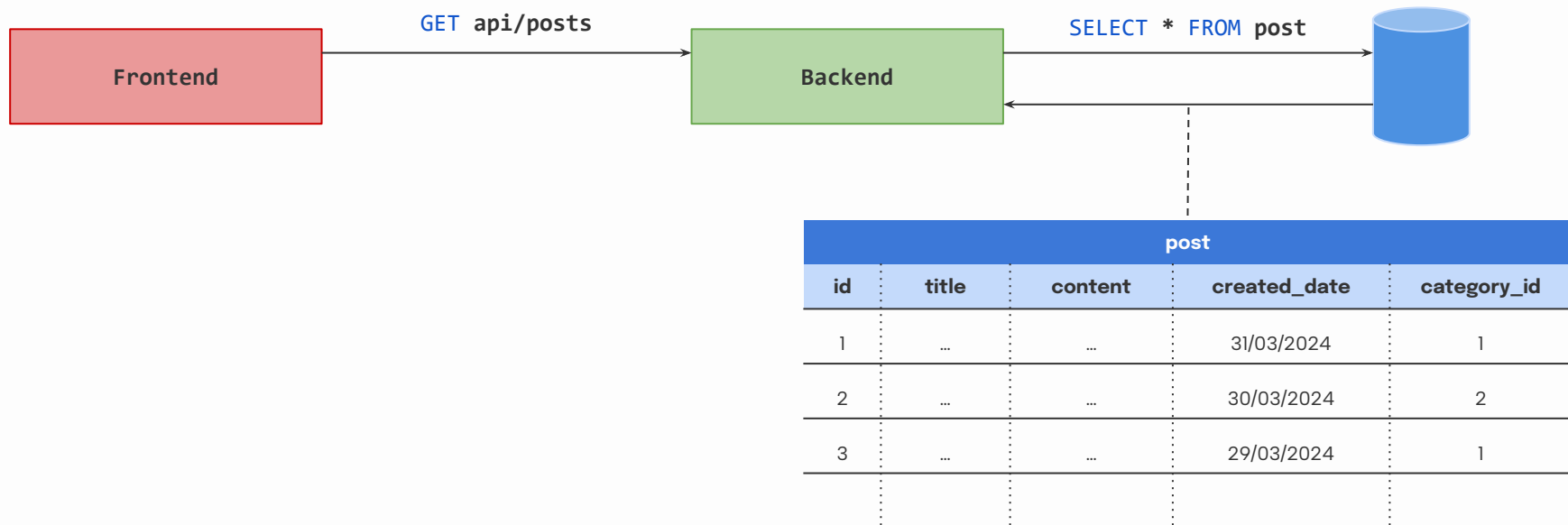
Flow frontend <-> backend



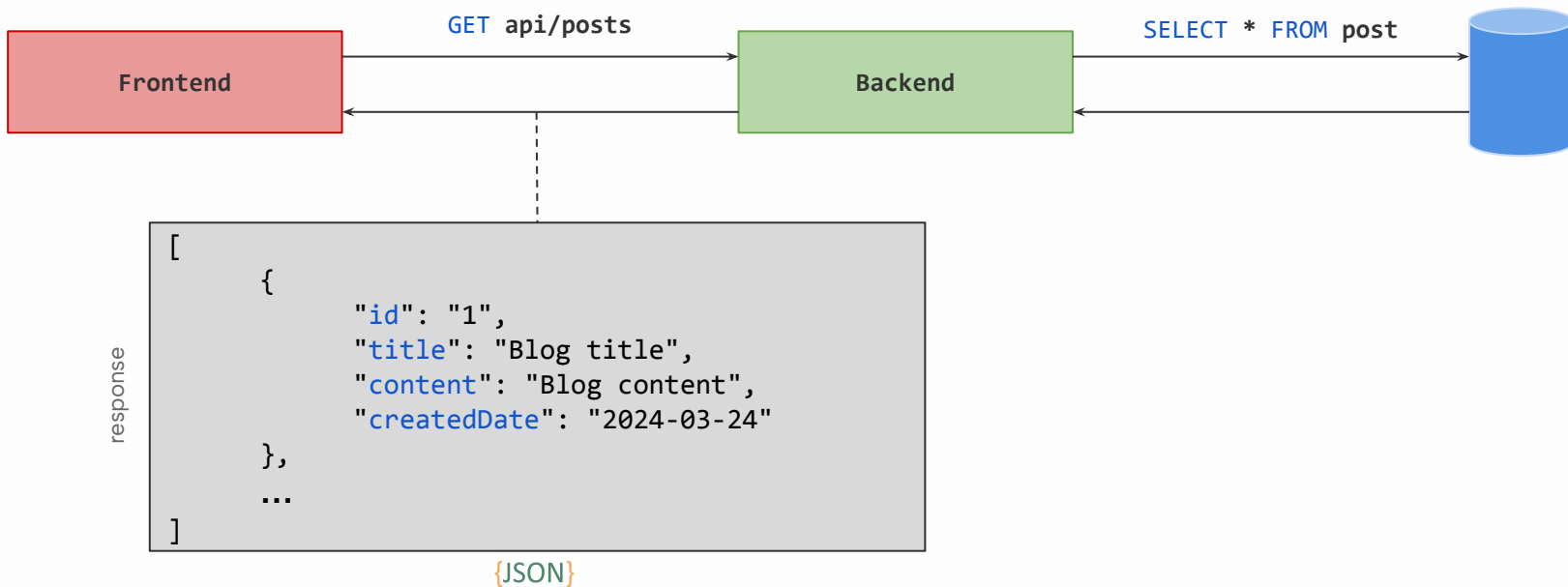
Flow frontend <-> backend



Flow frontend <-> backend



Flow frontend <-> backend

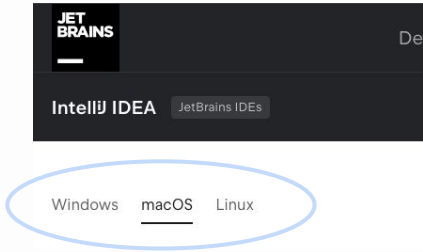


05

Before next session



Download IntelliJ IDEA as your Java IDE

Step 1	Head over to jetbrains.com/idea/download	
Step 2	Select your OS (Windows, macOS or Linux)	
Step 3	Download IntelliJ IDEA Community Edition (free)	